

Anticoagulation Timing After AF-Related Stroke

The Clinical Dilemma: Competing Risks

Early Ischemic Risk

High risk of recurrent stroke (up to 1–2% per day) in first 0–14 days

Recurrence often disabling or fatal

Drives the need for **early** anticoagulation

Early Hemorrhagic Risk

Infarcts are prone to hemorrhagic transformation (HT)

Risk highest in large infarcts and early time points

Drives the fear of **early** anticoagulation



Balancing Competing Risks

Evolution of Evidence

IPD Meta-Analysis (CATALYST)

Individual RCTs (ELAN, TIMING, etc)

Observational Studies (e.g., RAF Study)

Expert Opinion / Dogma (e.g., "1-3-6-12 Rule")

Four RCTs

Study (N)	Primary Outcome (Time)	Design	Early Arm Protocol	Late Arm Protocol	Primary Result (Early vs. Late)
TIMING (888)	Rec. Stroke, sICH, All-Cause Mort. (90d)	Noninferiority Margin: 3%	≤4 days (Pragmatic)	5–10 days (Pragmatic)	6.89% vs 8.68% (NI Met)
ELAN (2013)	Rec. Stroke, Sys. Emb., Major EC Bleed, sICH, Vasc. Death (30d)	Estimation design	Imaging-based (≤48h or 6–7d)	Minor: days 3-4 Moderate: 6-7 Major: 12-14	2.9% vs 4.1% (p=0.14)
OPTIMAS (3621)	Rec. Stroke, sICH, Unclass. Stroke, Sys. Emb. (90d)	Non-Inferiority (2.0%)	≤4 days (Pragmatic)	7–14 days (Pragmatic)	3.3% vs 3.3% (NI Met)
START (200)	Ischemic or Hemorrhagic Event (30d)	Adaptive Bayesian	Day 3–4	Day 6, 10, or 14	Composite counts: 1 / 4 / 3 / 2 (days 3-4 / 6 / 10 / 14)

CATALYST: Primary Outcome (30 Days)

Primary Composite: Rec. Ischemic Stroke, sICH, or Unclassified Stroke

Early ($\leq 4d$): 2.1% (57/2683) vs. Later ($\geq 5d$): 3.0% (83/2746)

Absolute Risk Reduction (ARR)

0.9%

NNT: 111

Relative Risk Reduction (RRR)

30%

OR: 0.70 (95% CI 0.50–0.98; $p=0.039$)

CATALYST: Key Efficacy & Safety Outcomes (30-Day)



Benefit driven by ischemic stroke reduction ($p=0.029$). No significant difference in sICH ($p=0.96$) or mortality ($p=0.19$)

Clinical Application: Exclusions & Protocols

Trial-specific scope

Eligibility and imaging exclusions differ by trial. Fixed thresholds of $>2/3$ MCA territory or BP $>185/110$ are not a universal exclusion rule.

OPTIMAS excluded PH2 and planned surgery requiring interruption or delay of anticoagulation during the first week. START allowed selected hemorrhagic transformation.

Trial timing: TIMING/OPTIMAS early within 4 days; ELAN early within 48 hours for minor/moderate stroke and days 6-7 for major stroke. These are trial strategies, not a universal treatment algorithm.

Parenchymal hemorrhage requires separate caution; the ELAN HT analysis reported possible worse functional outcome with early treatment.

CATALYST: Pooled Patient Profile (N=5,441)

Mean age	77.7 (SD 10.0)
Median NIHSS	5 (IQR 3–10)
NIHSS Distribution	0–4 (Mild): 42.8%
	5–10 (Moderate): 32.1%
	≥11 (Severe): 24.4%
On OAC at Onset	32.7%
Reperfusion Therapy	32.5%

ELAN HT analysis: central imaging included PH2. This is not a pooled CATALYST baseline variable or blanket subgroup-safety conclusion. See Repair 04.

Nuance: 30-Day vs. 90-Day Primary Outcome

Primary Outcome at 30 Days

OR 0.70 (95% CI 0.50–0.98)

p = 0.039

Statistically significant reduction

Primary Outcome at 90 Days

OR 0.88 (95% CI 0.65–1.19)

p = 0.40

No significant difference

- *Benefit of early initiation is "front-loaded" in the highest-risk period*
- *After acute phase, both groups are on effective prevention, diluting the treatment effect over time*

Nuance: Functional Outcomes (90-Day mRS)

ELAN (N=2,013)

ELAN: specify the timepoint

mRS 0-2: 66.6% early vs 65.8% later at 90 days.

The 62.6% / 62.6% pair belongs to 30 days.

OPTIMAS (N=3,621)

OPTIMAS: source gap

The cited main report deferred functional outcomes to another publication. The displayed percentages and ordinal OR were not located in its main report or supplement.

These sources do not support a combined claim that both trials showed no difference in 90-day disability. Functional outcome and recurrent-stroke prevention are distinct endpoints.

Nuance: Subgroup Consistency (p-interaction)

Read subgroup results with their limits

NIHSS: CATALYST interaction $p=0.54$ compared four strata. It was not a dedicated safety test in severe stroke.

Prior OAC: the reported interaction $p=0.89$ does not prove absence of harm.

Hemorrhagic transformation: no HT interaction $p=0.61$ was reported in the cited CATALYST main report/supplement; that analysis was future work. ELAN HT results require separate interpretation, including possible worse functional outcome with early treatment in PH.

Nonsignificant interaction tests do not establish equal effects or safety in each subgroup.

Radiographic Subgroups

ELAN central imaging identified HI and PH, including PH2. Its HT analysis raised possible functional harm with early treatment in PH. CATALYST did not report the asserted HT $p=0.61$; its main report deferred HT analyses. The ELAN HT correction body remains unread.



- **Infarct Size:**

ELAN substudy: Analyzed outcomes by radiographic size criteria

Benefit of early initiation consistent across Minor, Moderate, and Major infarcts (p -interaction=0.81)

Source limit: the displayed infarct-size interaction $p=0.81$ was not paired to the cited report in this repair. It should not be used as proof of subgroup safety.

Methods

Trial-Level Data & Meta-Analysis Methods

Trial-Specific: Definitions of Severity

Trial	Method	Mild	Moderate	Severe / Major
ELAN	Imaging-Based (per text)	Infarct $\leq$ 1.5 cm	Infarct in distribution of a cortical superficial branch of MCA, ACA, or PCA	Larger infarcts in these arteries, or brain-stem/cerebellar infarct $>$ 1.5 cm
OPTIMAS	NIHSS-Based (Stratification)	0-4	5-10 / 11-15	16-21 / $>$ 21
TIMING	NIHSS-Based (Subgroups)	0-3	4-5 / 6-10	11-15 / $>$ 15
START	START: infarct $>$ 50% MCA was excluded. NIHSS $>$ 23 was an exclusion when lesion extent was not visible on imaging; it was not an unconditional NIHSS cap.			

Trial Analysis: ELAN (N=2,013)

Design & Population

Estimation design; no prespecified superiority or noninferiority hypothesis.

Early: within 48 hours for minor/moderate stroke; days 6-7 for major.

Later: days 3-4 minor, 6-7 moderate, 12-14 major.

Admission NIHSS median 5 (IQR 2-11); randomization median 3. No universal <100-hour enrollment rule established.

Protocol PH exclusions did not prevent centrally identified PH in the HT analysis; interpret that analysis separately.

30-Day Outcomes

Outcome	Early (N=1006)	Late (N=1007)	OR (95% CI); p-value
Primary Composite	2.9%	4.1%	0.70 (0.44-1.14); p=0.14
Recurrent Ischemic Stroke	1.4%	2.5%	0.57 (0.29-1.07)
Symptomatic ICH	0.2%	0.2%	1.02 (0.16-6.59)

Trial Analysis: TIMING (N=888)

Design & Population

Pragmatic noninferiority trial; margin 3 percentage points. Randomization within 72 hours; early DOAC within 4 days versus days 5-10.

Median admission NIHSS 4 (IQR 2-8).

Eligibility depended on suitability for a DOAC; do not infer a universal PH2 or infarct-volume cutoff from the slide.

Recruitment stopped at 888 of 3,000 planned; severe-stroke and rare-bleeding inferences remain limited.

90-Day Outcomes

Outcome	Early (N=450)	Late (N=438)	Risk Diff (95% CI); p-value
Primary Composite	6.89%	8.68%	-1.79% (-5.31 to 1.74); p(NI)=0.004
Recurrent Ischemic Stroke	3.11%	4.57%	-1.46% (-3.98 to 1.07)
Symptomatic ICH	0.0%	0.0%	N/A

Trial Analysis: OPTIMAS (N=3,621)

Design & Population

Pragmatic noninferiority trial; margin 2 percentage points. Early DOAC within 96 hours versus days 7-14. Approach within 72 hours was encouraged, not a universal treatment cutoff.

Excluded PH2 and surgery already planned requiring anticoagulant interruption or delay during week 1. A fixed >2/3 MCA cutoff is not established.

Few very severe strokes and only 23 symptomatic ICH events limit precision. The trial does not identify an optimal day within either treatment window.

90-Day Outcomes

Outcome	Early (N=1814)	Late (N=1807)	Adj. Risk Diff (95% CI); p-value
Primary Composite	3.3%	3.3%	0.000 (−0.011 to 0.012); p(NI)=0.0003
Recurrent Ischemic Stroke	2.4%	2.3%	−0.001 (−0.011 to 0.009)
Symptomatic ICH	0.6%	0.7%	0.001 (−0.004 to 0.006)

Trial Analysis: START (N=200)

Design & Population

- **Design:** Adaptive Bayesian (to find *optimal* time)
 - **Population:** AF-stroke; Median NIHSS=6.5 (IQR 4-14)
Selected HT could be included. Excluded >50% MCA infarction, or NIHSS >23 when lesion extent was not visible.
 - **Intervention:** Multiple arms
 - Day 3-4 vs. Day 6 vs. Day 10 vs. Day 14
 - **Limitations:**
 - Terminated early (N=200); low power
 - High loss to follow-up (10% at 30d)
 - Potential selection bias (slow enrollment)
- Selected HT was allowed; this does not establish safety in every HT phenotype.

30-Day Primary Outcome

Arm	Composite events
Day 3-4 (N=54)	Day 3-4: 1 / 54
Day 6 (N=53)	Day 6: 4 / 53
Day 10 (N=46)	Day 10: 3 / 46
Day 14 (N=47)	Day 14: 2 / 47

Methods

Statistical Model

- One-stage IPD meta-analysis
- Generalized linear mixed-effects model
- Random effect for trial (handles clustering)

Primary ORs unadjusted for patient covariates.

Trial Design & Adjudication

Trial-specific outcome ascertainment; no new central blinded readjudication of all outcomes. Table S1 reports no TIMING event adjudication. Endpoint definitions differed.

Data Harmonization

- ****Early Arm:**** Initiation ≤ 4 days
- ****Later Arm:**** Initiation ≥ 5 days
- Harmonized split based on TIMING & OPTIMAS pragmatic design

CATALYST: Statistical Robustness

Separate the primary result from sensitivity analyses

Aggregate-data sensitivity (Supplement Figure S2):

Common-effect OR 0.713 (95% CI 0.505-1.005).

Random-effects OR 0.721 (95% CI 0.510-1.020).

Both intervals include 1.

The exact per-protocol and leave-one-out estimates previously displayed were not located in the recovered main report/supplement. They are withheld as unverified, not classified as proven false.

Major Methodological Biases & Confounders

Open-Label Design (All Trials)

Outcome ascertainment differed by trial; no universal central blinded adjudication.

Protocol Heterogeneity (Meta-Analysis)

Combining dissimilar trial designs (imaging-based vs. pragmatic)

Varied control arm definitions (e.g., 5-10d, 7-14d, 12-14d)

Antiplatelet use during delay: part of the compared strategy; direction of effect on the composite is not established.

Generalizability: exclusions and enrolled populations differed. PH and very large strokes require trial-specific interpretation; selected HT was enrolled.

Early Termination (TIMING, START)

Trials were underpowered; risk of unstable/biased effect estimates

Methodological Note: Antiplatelet "Bridging"

 **Heterogeneous Protocols:** No uniform protocol for antiplatelet (AP) use in the "gap" period

Randomization compared early versus delayed DOAC strategies. Antiplatelet use during the delay was a co-intervention, not automatically confounding of the randomized contrast.

Its direction of effect on a composite of ischemic and bleeding events was not established. The cited evidence does not support a definite bias toward the null.

Methodological Note: DOAC Choice & Dosing

 **Pragmatic Choice:** Trials did not mandate a specific DOAC

TIMING (N=888): Apixaban (55%), Dabigatran (28%), Rivaroxaban (13%), Edoxaban (3%)

OPTIMAS (N=3621): Apixaban (62%), Edoxaban (29%), Rivaroxaban (5%), Dabigatran (2%)

OPTIMAS: 816 received reduced doses; 155 (19.0%) were classified underdosed by NICE guidance (Table e1).

Source gap: the previously displayed drug-choice interaction $p=0.37$ was not located in the recovered main report/supplement.

Remaining Uncertainties & Future Directions

PH-2 & Malignant Infarcts

PH2 was not universally absent: ELAN central imaging identified cases.

OPTIMAS excluded PH2 and certain planned surgery.

Timing in these high-risk settings remains uncertain; avoid blanket early-treatment reassurance.



Warfarin Patients

- Data is DOAC-only
- No evidence for patients requiring Warfarin (e.g., mechanical valves, APLAS, severe CKD)



Septic Embolism

- Evidence applies **only** to AF-related stroke
- **Cannot** be applied to suspected septic embolism from endocarditis



Future: < 24 Hours?

- "Early" arm (≤ 4 days) does not define optimal time **within** that window
- ****ASAP Trial (NCT06057467):****
Ongoing RCT testing 0-24h vs. 3-4 days in mild-moderate stroke (NIHSS 0-15)

Totality of Evidence

CATALYST (5,441 participants) favored early DOAC for its 30-day composite of recurrent ischemic stroke or symptomatic ICH. The roughly 30% relative reduction refers to that composite; recurrent ischemic stroke OR was 0.66.

This does not establish safety for every HT phenotype. Trial-specific selection and the ELAN PH functional-outcome concern limit generalization.

Targeted repairs | 30 September 2026

The original deck is an archive. These numbered notes document targeted corrections; unchanged material has not been comprehensively updated. Trial selection, source conflicts and unread correction notices remain explicit. Original slide numbers are preserved.

Repair 01 - TIMING: noninferiority margin

TIMING used a 3 percentage-point noninferiority margin. Early treatment was within 4 days; randomization occurred within 72 hours. The indexed TIMING correction body remains unavailable; this repair is limited to the paired original report.

TIMING | <https://pmc.ncbi.nlm.nih.gov/articles/PMC9648987/>

Repair 02 - ELAN: study design and treatment windows

ELAN used an estimation design without a prespecified superiority or noninferiority hypothesis. Early DOAC: within 48 hours for minor/moderate stroke, days 6-7 for major stroke. Later: days 3-4 for minor, 6-7 for moderate, 12-14 for major. Overall admission NIHSS was median 5 (IQR 2-11); the randomization NIHSS median was 3. A universal enrollment cutoff of 100 hours is not established by the cited main report. The recovered PDF declares an update on June 29, 2023; the update substance was not independently reconciled.

ELAN | <https://pubmed.ncbi.nlm.nih.gov/37222476/>

ELANprotocol | <https://pmc.ncbi.nlm.nih.gov/articles/PMC9720853/>

ELANfull | <https://eprints.gla.ac.uk/299352/2/299352.pdf>

Repair 03 - START: eligibility and composite events

START allowed selected hemorrhagic transformation at investigator discretion. Exclusions included infarct volume greater than half the MCA territory, or NIHSS greater than 23 if the lesion extent was not visible. For days 3-4, 6, 10 and 14, ischemic events were 0/3/2/2 and hemorrhagic events 1/1/1/0: the composite counts were 1/4/3/2. Randomized arm sizes were 54/53/46/47. The trial did not identify a clearly superior start day.

START | <https://jamanetwork.com/journals/jamaneurology/fullarticle/2831918>

Repair 04 - Hemorrhagic transformation: limits of reassurance

Hemorrhagic transformation is heterogeneous. The ELAN analysis included centrally identified PH2 despite protocol exclusions and reported possible worse functional outcome with early treatment in patients with parenchymal hemorrhage. Nonsignificant interaction tests do not establish subgroup safety. The ELAN HT erratum (PMID 38950115) was identified but its body remains unread; the repair therefore withholds blanket reassurance rather than declaring corrected-current subgroup clearance.

ELANHT | <https://pubmed.ncbi.nlm.nih.gov/38753452/>

START | <https://jamanetwork.com/journals/jamaneurology/fullarticle/2831918>

Repair 05 - CATALYST: mean age

CATALYST reported mean age 77.7 years (SD 10.0), not median age. The pooled sample comprised 5,441 participants.

CATALYST | <https://pubmed.ncbi.nlm.nih.gov/40570866/>

Repair 06 - Trial-specific eligibility and timing

Eligibility must be interpreted separately for each trial. A universal exclusion list using infarct greater than two-thirds MCA territory, blood pressure 185/110, or all PH2 is not established across these trials. OPTIMAS excluded PH2 and a decision already made for surgery requiring anticoagulant interruption or delay during the first week; the protocol encouraged approach within 72 hours, but the early strategy allowed treatment within 96 hours. These trial windows do not constitute a universal bedside algorithm.

TIMING | <https://pmc.ncbi.nlm.nih.gov/articles/PMC9648987/>

OPTIMASregistry | <https://clinicaltrials.gov/study/NCT03759938>

ELANprotocol | <https://pmc.ncbi.nlm.nih.gov/articles/PMC9720853/>

OPTIMASfull | <https://eprints.soton.ac.uk/497252/1/1-s2.0-S0140673624021974-main.pdf>

OPTIMASsupp | <https://ars.els-cdn.com/content/image/1-s2.0-S0140673624021974-mmc1.pdf>

Repair 07 - Antiplatelet co-interventions and causal interpretation

The randomized comparison was an early versus delayed DOAC strategy. Antiplatelet use during the delay is a co-intervention, not automatically a failure of randomization. Its direction of effect on a composite containing ischemia and bleeding was not established by the cited trials; a claim of bias toward the null has been removed.

START | <https://jamanetwork.com/journals/jamaneurology/fullarticle/2831918>

TIMING | <https://pmc.ncbi.nlm.nih.gov/articles/PMC9648987/>

Repair 08 - CATALYST: composite versus ischemic-stroke effect

In CATALYST, the roughly 30% relative reduction describes the primary 30-day composite. Recurrent ischemic stroke had an odds ratio of 0.66, with an approximately 34% crude relative risk reduction. The pooled finding is not proof of safety in every hemorrhagic-transformation phenotype.

CATALYST | <https://pubmed.ncbi.nlm.nih.gov/40570866/>

Repair 21 - Functional outcomes: timepoints and source gaps

ELAN Table 2 reports mRS 0-2 in 62.6%/62.6% at 30 days and 66.6%/65.8% at 90 days (early/late). OPTIMAS main report explicitly deferred functional outcomes to another publication; its recovered main report and supplement did not support the previously shown functional percentages or ordinal odds ratio. For archival provenance only, the unverified OPTIMAS values were 52.8%/52.5%, ordinal OR 1.00 (0.90-1.12), $p=0.96$. They are not validated results; their source remains unresolved. OPTIMAS PDF declares a January 2, 2025 correction; the whole notice remains unread.

ELANfull | <https://eprints.gla.ac.uk/299352/2/299352.pdf>

OPTIMASfull | <https://eprints.soton.ac.uk/497252/1/1-s2.0-S0140673624021974-main.pdf>

OPTIMASsupp | <https://ars.els-cdn.com/content/image/1-s2.0-S0140673624021974-mmc1.pdf>

Repair 22 - CATALYST: statistical model and adjudication

CATALYST used a one-stage generalized linear mixed-effects model with a trial random effect; its primary odds ratios were unadjusted for patient-level covariates. Trial-specific endpoint ascertainment was retained, rather than new central blinded readjudication of all outcomes. Supplement Table S1 reports no event adjudication for TIMING and differing endpoint definitions.

CATALYSTfull | <https://lirias.kuleuven.be/retrieve/b905732a-875d-418c-bb75-fc8228d8e9ab>

CATALYSTsupp | <https://ars.els-cdn.com/content/image/1-s2.0-S0140673625004398-mmc1.pdf>

Repair 23 - CATALYST: subgroup interaction limits

CATALYST Figure 3 reports NIHSS interaction $p=0.54$ across four strata, not a dedicated safety test for NIHSS at least 11. The cited main report and supplement did not report an HT interaction $p=0.61$; HT analysis was described as future work. No absence-of-harm inference should be made from a nonsignificant interaction. ELAN HT evidence has separate limitations in Repair 04.

CATALYSTfull | <https://lirias.kuleuven.be/retrieve/b905732a-875d-418c-bb75-fc8228d8e9ab>

CATALYSTsupp | <https://ars.els-cdn.com/content/image/1-s2.0-S0140673625004398-mmc1.pdf>

Repair 24 - CATALYST: sensitivity analyses and source gaps

CATALYST supplement Figure S2 gives a common-effect OR 0.713 (95% CI 0.505-1.005) and random-effects OR 0.721 (0.510-1.020). Both confidence intervals include 1. The previously asserted significant aggregate sensitivity result has been corrected. Exact per-protocol and leave-one-out estimates on the original slide were not located in the recovered main report/supplement; they remain unverified, not proven false. Archival provenance only: per-protocol $n=5,110$, OR 0.69 (0.49-0.98), $p=0.037$; leave-one-out ORs without TIMING 0.69 (0.48-0.99), without ELAN 0.83 (0.54-1.28), without OPTIMAS 0.61 (0.39-0.97), without START 0.70 (0.49-0.99). These exact figures are not validated by the sources acquired.

CATALYSTfull | <https://lirias.kuleuven.be/retrieve/b905732a-875d-418c-bb75-fc8228d8e9ab>

CATALYSTsupp | <https://ars.els-cdn.com/content/image/1-s2.0-S0140673625004398-mmc1.pdf>

Repair 25 - OPTIMAS: reduced dosing and concordance

OPTIMAS supplement Table e1 reports 816 reduced-dose recipients: 378 apixaban, 21 dabigatran, 369 edoxaban, 48 rivaroxaban. Of these, 155 (19.0%) were classified as underdosed using NICE guidance (88/4/52/11). Reduced dosing was therefore not uniformly guideline-concordant. The originally displayed drug-choice interaction $p=0.37$ was not located in the recovered sources and is not treated as validated.

OPTIMASfull | <https://eprints.soton.ac.uk/497252/1/1-s2.0-S0140673624021974-main.pdf>

OPTIMASsupp | <https://ars.els-cdn.com/content/image/1-s2.0-S0140673624021974-mmc1.pdf>